

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6.	(a)	(i)		Voltage	1			1		1
		(ii)		Current	1			1		1
		(iii)		Scale (1) Plot ($\pm < 1$ small square) (2) smooth curve (1)		4		4	4	4
		(iv)		Correct reading from graph = 1.4 approx. (1) 7.2/1.4 (ecf) (1) 5.14 (1)	1	2		3	3	3
		(v)		At lower voltages the graph is steeper (1) So resistance is 1/gradient (1) So resistance is lower at the start (1) OR At lower voltages the graph is straight (1) Resistance is constant (1) Then curves so resistance increases (1) OR Calculation of two resistances may be shown for 2 marking points + statement for 3 rd mark (As current increases) temperature of the lamp increases (1) Increasing the resistance (1) Non linear (1)	1	2		3		3

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	(b)			Any 4 × (1) from: Resistance for resistor is constant (for all voltages) (1) Resistance of a diode changes / decreases (1) The resistance of the diode is {very large/infinite} for negative voltages (1) The resistance of the diode is {very large/infinite} for small positive voltages (1) as the voltage reaches a particular level the resistance decreases / (accept: as the voltage increases the resistance decreases) (1)	2	2		4		4
				Question 6 total	6	10		16	7	16